

CASE FILE 0814

The Broken Alibi

Blackwood Technologies HQ | August 14, 22:00 to 00:00

Investigators: Abdalla, Dante, Kunal



CONFIDENTIAL





Three statements. One silence.



EMP - 0047

"On the 3rd floor working on the audit files. Left around midnight."



EMP - 0031

"In the server room fixing a batch job. Left before 11 PM."



EMP - 0092

"Came back to grab something from my desk. I was in and out."



EMP - 0011

No statement on record. The data puts them in the building.

**PERSON OF
INTEREST**



Three systems that never talk

EXHIBIT A



Badge access

Door swipes, in and out.

Arrives once a night as a file

EXHIBIT B



Device logs

Logins, file reads, USB, email.

Streams in line by line

EXHIBIT C



Transactions

Cafeteria, vending, parking.

Live payment database

Three formats. Three delivery speeds. Three clocks that disagree.

How evidence moves through our lab

1 SOURCES



Badge access

Nightly CSV,
door codes



Device logs

Streaming
syslog,
key=value



Transactions

Live
PostgreSQL,
terminals

2 INGESTION

Nightly batch

Checks the vendor's
hash, then hashes
each row

Stream consumer

Hashes each line,
commits its offset

Change capture

Reads the change
log, never the live
table

3 CLEAN

Raw + hashed records

Clean & normalize

- Times to UTC, original kept
- One ID per person
- event= to event_type
- Door codes to rooms
- IPs normalized
- Duplicates removed
- Bad lines quarantined
- Clock gaps flagged, not fixed

4 STORAGE

RAW ZONE (write-once)

Exact bytes as received,
hashed per file and per record.
Production: S3 Object Lock

CURATED ZONE

One timeline table. Each row
keeps its raw_event_hash.
Rebuilt from raw every run.
Production: Iceberg / BigQuery

AUDIT LOG

Hash-chained. Records every
load, clean and query.

5 SERVING

Timeline + alibi SQL

KEY CONTROL

Re-hashes every
answer row against
raw before it is shown

Production:
investigation portal

Outputs

timeline.csv
alibi_findings.csv
custody_report.json



Three calls we'd defend in court

1

Keep the original, clean a copy

Costs double storage. Buys proof: any answer can be traced back to untouched evidence.

2

Flag the clocks, don't fix them

Parking gate and badge reader disagree by up to 8 min 13 s. The gap changes per person, so any correction is a guess. We mark events whose order is uncertain.

3

Quarantine, don't drop

A line we can't read is kept with its error. No evidence disappears quietly.

One question, asked of every system

```
SELECT event_timestamp, employee_id,  
       event_source, description,  
       raw_event_hash  
FROM curated.employee_activity_timeline  
WHERE employee_id IN ('EMP-0047', 'EMP-0031',  
                     'EMP-0092', 'EMP-0011')  
      AND event_timestamp BETWEEN  
          '2026-08-14 21:45' AND '2026-08-15 00:15'  
ORDER BY event_timestamp;
```

Simplified from sql/01_timeline.sql

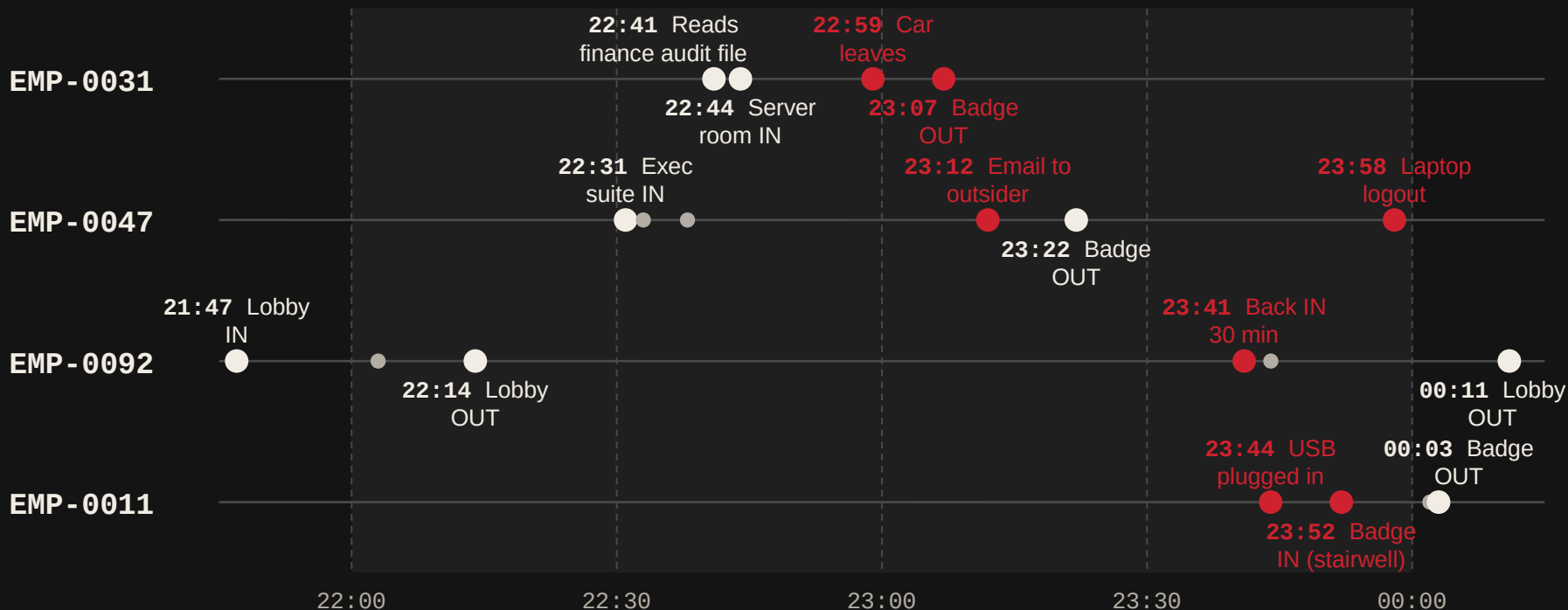
16

events inside 22:00 to 00:00

20

with a 15-minute margin each
side

August 14, minute by minute



Red = contradicts a statement or needs explaining. Shaded band = 22:00 to 00:00.

One lie, two cracks, one shadow

EMP - 0031

CONTRADICTED

Said gone before 23:00. Badge out of the server room at 23:07:44. Also read the finance audit file at 22:41.

EMP - 0047

DOESN'T HOLD

Badged out at 23:22, laptop active until 23:58, no building exit recorded. Emailed an outside address at 23:12.

EMP - 0092

DOESN'T HOLD

"In and out" was two visits of 28 and 30 minutes. The second ended at 00:11.

EMP - 0011

PERSON OF INTEREST

USB at 23:44:17, badge in at 23:52:07. A 7 min 50 s gap is too big for clock drift. Got in without badging, or someone used their machine.



Nobody touched the evidence

EVIDENCE | DO NOT TAMPER

3 / 3 source files match the published hashes

20 / 20 raw records match the hash taken at intake

20 / 20 answer rows re-hashed against the raw record

25 audit log entries, chain intact



We tried to break it

Moved EMP-0031's exit from 23:07:44 to 22:57:44 to fit their alibi.

File, line and timeline row all failed the check. The pipeline refused to rebuild.

**SEAL
UNBROKEN**

CASE STATUS

Open. Leads identified.



Question first: EMP-0011, then EMP-0031

2 hours rebuilt | 4 statements tested | 0 records altered

```
$ python run_pipeline.py --reset
```

Anyone can rerun the full investigation and get the same answer.

QUESTIONS?